Experiment 2: Bayes Decision Theory

Aim

To classify SMS messages as Spam or Ham using the Naive Bayes algorithm.

Tools Required

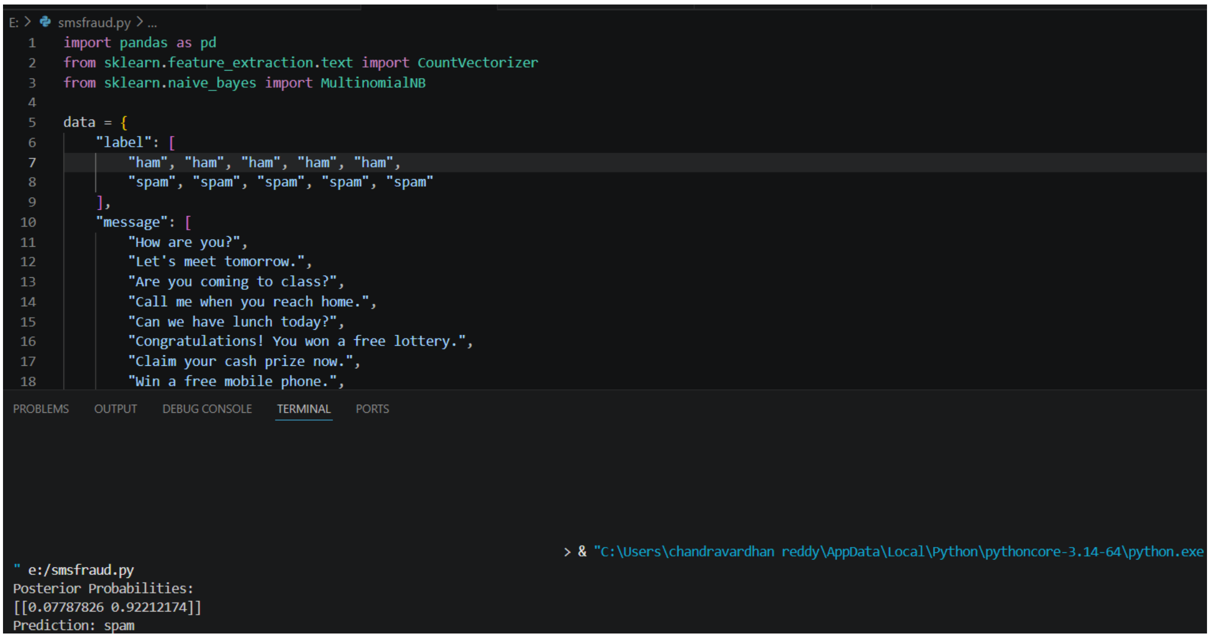
- Python
- Pandas
- VS Code

Code

```
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
data = {
    "label": [
        "ham","ham","ham","ham","ham",
        "spam","spam","spam","spam","spam"
    ],
    "message": [
        "How are you?",
        "Let's meet tomorrow.",
        "Are you coming to class?",
        "Call me when you reach home.",
        "Can we have lunch today?",
        "Congratulations! You won a free lottery.",
        "Claim your cash prize now.",
        "Win a free mobile phone.",
        "Exclusive offer! Click here.",
        "You have won a $1000 gift card."
    ]
}
df = pd.DataFrame(data)
vectorizer = CountVectorizer()
X = vectorizer.fit_transform(df["message"])
```

```
y = df["label"]

model = MultinomialNB()

model.fit(X, y)

test_message = ["Congratulations! You won a free lottery ticket"]

test = vectorizer.transform(test_message)

print("Posterior Probabilities:", model.predict_proba(test))

print("Prediction:", model.predict(test)[0])
```

Output



Result

Bayes Decision Theory was successfully implemented using the Naive Bayes classifier, and the SMS message was correctly classified as Spam.